

STRICT TECHNICAL EVALUATION · COMPETITION JUDGE REVIEW

PredicTrace

Predictive Cash-out Intelligence for Cybercrime Complaints — SIH26184

FINAL VERDICT: ALMOST READY

Overall score: 74 / 100

Scope reviewed: Full frontend, backend, ML pipeline, graph engine, data generator, tests, evaluation artifacts

Method: Traced actual implementation file-by-file — no claim below is taken from a filename or docstring alone

Part 1 — Overall Score

DIMENSION	SCORE	WHY
Problem relevance	9/10	Cash-out interdiction is a real, named gap in cybercrime response (money moves faster than paperwork). Framing is precise, not generic "AI for good."
Technical depth	8/10	Real graph construction, real Louvain, real point-in-time leakage discipline, real calibration math. Depth is genuine where it exists.
Backend quality	8/10	Clean separation (graph_engine / ml / api / core), shared <code>pipeline.py</code> so train and eval can't silently diverge, incremental caching engineered under real measured latency pressure. Some modules (activity simulator) are patched onto the side rather than designed in from the start.
ML quality	8/10	Legitimate binary-classification-as-ranking setup, complaint-level (not row-level) calibration split, FrozenEstimator+CalibratedClassifierCV, SHAP wired to the raw base model correctly. Single 80/20 temporal split, no cross-validation, one random seed reported.
Graph intelligence	7/10	Graph features demonstrably feed the model and produce a measured lift. Community detection uses the network's own Louvain — real, not GNN-dressed marketing. Under-visualized relative to how much work went into building it correctly.
Explainability	7/10	SHAP values are real per-row values from the actual fitted booster — not decorative. Narrative sentence generation is templated, not SHAP-native.
Cross-case intelligence	8/10	Genuinely does $A \rightarrow X \rightarrow Y$ / $B \rightarrow Z \rightarrow Y$ reasoning through discovered community membership, not a lookup table. Strongest real differentiator in the project.
Frontend/UI	6/10	Data density is real, but the visual language is a generic dark-admin-dashboard, and Command Center has accumulated widgets across many iterations without a hard edit pass.
Investigator usability	6/10	The single most actionable screen (Approve/Reject) was originally buried in the last of 4 tabs and had to be manually pulled to the persistent header — that it needed manual correction is itself a usability signal.

Data realism	7/10	Real named localities, realistic hop/delay/fee structure, an explicit held-out "unseen pattern" generalization set. Still obviously templated narrative text and a small (690-complaint) universe.
Evaluation quality	9/10	A real, held-out, temporal-split, baseline-vs-advanced report — actually computed, actually stored as JSON, actually regression-tested. Rare at this level.
Real-world feasibility	4/10	Demo-mode auth, in-memory-only settings/event log, SQLite, no audit trail, no drift monitoring — explicitly disclosed, but still a real gap.
Innovation/differentiation	7/10	Lift-over-random-chance urgency framing, point-in-time discipline, city-scoped RBAC-in-data are more thoughtful than typical hackathon builds.
Demo impact	7/10	Simulate Complaint → real prediction → real explanation → real cross-case link is genuinely impressive live. The background Live Feed simulator is a risk if a judge asks the wrong question.
Engineering quality	8/10	Leakage-guard test, evaluation-floor regression test, shared train/eval pipeline, careful cache invalidation — unusually disciplined for the category.

74/100

NOT A SHELL. NOT VAPORWARE. A REAL, MOSTLY-HONEST SYSTEM.

Overall: a real, mostly-honest system with a genuinely computed ML core, sitting under a UI that hasn't had a final edit pass and a "live feed" feature whose honesty depends entirely on how it's explained (see Part 11).

Part 2 — "Is This Actually Intelligent?"

COMPONENT	CLASSIFICATION	EVIDENCE
Entity extraction	REAL, but shallow	<code>nlp/entity_extraction.py</code> — real regex/substring matching, not hardcoded output. But it's a 6-bank hardcoded list + two regexes, working only because narrative templates embed the bank name/IFSC verbatim. Real string processing, not real NLP — the code's own docstring already admits this.
Graph construction	REAL COMPUTATION	<code>build_graph.py</code> builds an actual networkx MultiDiGraph from real rows, with genuine point-in-time filtering.
Community detection	REAL COMPUTATION	Real Louvain (<code>louvain_communities</code>) on a real weighted projection. Verified separate from generator's ground-truth <code>ring_id</code> by an automated leakage test.
Centrality	REAL, but thin	Real networkx degree/betweenness calls, but betweenness only computes on communities ≥ 3 members — weak signal for small communities.
Cross-case linking	REAL COMPUTATION	Derived from actual shared account membership within a Louvain community, cross-referenced against real transaction/withdrawal rows.
Temporal features	REAL COMPUTATION	Recency decay computed from the real last-known transaction timestamp relative to the real <code>filed_at</code> .
Geographic features	REAL COMPUTATION	Real haversine-kernel density over real historical withdrawal-point coordinates, self-excluding the complaint's own event.
ML prediction	REAL COMPUTATION	XGBoost <code>predict_proba</code> on real fused features. Verified via the actual <code>evaluation_report.json</code> : baseline $P@5 = 12.3\%$, advanced = 48.6%.
Top-5 ranking	REAL COMPUTATION	Candidates sorted by real predicted probability within each complaint.
Confidence	REAL, calibrated	CalibratedClassifierCV output, not a raw score or UI-invented number.
Calibration	REAL COMPUTATION	FrozenEstimator + CalibratedClassifierCV, complaint-level split specifically to avoid a real identified leakage mode.
SHAP	REAL, correctly wired	TreeExplainer on the raw, uncalibrated booster — the correct approach. Values are per-row, genuinely computed.

Fraud-ring detection	REAL, unsupervised	Louvain communities of size ≥ 3 touching ≥ 1 complaint, computed live on every cache refresh.
"Live Feed" case injection	REAL DATA, SIMULATED CADENCE	Reveals a real held-back complaint (real DB write), reads real cached predictions — but the 3-4 minute pacing is an artificial loop, since there is no real external event stream behind a static dataset. Disclosed in code comments, NOT disclosed in the UI.
Simulated "decision" in Live Feed	SIMULATED, explicitly not written to DB	By design — correctly refuses to auto-mutate real case status, since a decision needs human judgment.

Bottom line: the core prediction pipeline is real, traceable, and reproducible. The one thing dressed up to look more "live" than it is is the Command Center activity feed's cadence — and only its cadence, not its content.

Part 3 — ML Review

1. **What is the model predicting?** P(this candidate withdrawal point is the true cash-out location for this complaint) — binary classification per (complaint, candidate) pair.
2. **Target variable:** `label` = 1 if `withdrawal_point_id` matches the complaint's actual recorded withdrawal event.
3. **Features:** Baseline: amount, hour-of-day, bank match, global geo density (4). Advanced: those plus chain depth, recency decay, degree centrality, community size, local betweenness, community complaint count, community geo density (11 total).
4. **Labels come from:** the generator's own recorded withdrawal event — real structural ground truth, not guessed.
5. **Is the dataset suitable?** For proving the method — yes. For real-world accuracy claims — no, and the project doesn't claim that.
6. **Data leakage?** No proven leakage found. Point-in-time construction + self-exclusion of a complaint's own row from historical aggregates, plus an automated test guarding against ground-truth `ring_id` / `account_type` re-entering as features.
7. **Train/test split correct?** Temporal (first 80% by `filed_at` = train) — correct methodology. Only one split is drawn.
8. **Temporal relationships respected?** Yes, consistently, including graph construction itself.
9. **Graph built with future information accidentally?** Not found — `graph_cache.py`'s key includes `as_of`, and every caller that matters passes the complaint's own `filed_at`.
10. **Is the model actually learning patterns?** Yes — the baseline sits near a geo-density-only heuristic (12.3% P@5); the advanced model's jump to 48.6% is the size of lift you'd expect if community/chain/recency signal is genuinely informative.
11. **Is XGBoost appropriate?** Yes, for ~11 features on a few thousand rows — reasonable, not over-engineered.
12. **Is the baseline meaningful?** Yes — deliberately deprived of graph/temporal features to isolate "does the graph help."
13. **Is calibration actually implemented?** Yes, correctly — complaint-level split, sigmoid method chosen for a documented, specific reason (isotonic collapsed with too few calibration samples — found and fixed, not theorized).
14. **Is SHAP correctly connected?** Yes — TreeExplainer on `advanced_raw`, the same fitted booster `advanced_calibrated` wraps.
15. **Are confidence values trustworthy?** As calibrated probabilities on this dataset — yes, Brier score (0.0236) is genuinely low. As a claim about real-world accuracy — no, and not claimed.
16. **Could the model exploit synthetic-data artifacts?** Almost certainly, partially — yes. Rings favor "hot" withdrawal points 70% of the time, and community geo-density/complaint-

count features are precisely engineered to detect exactly that regularity. Not cheating, but a fair thing a hostile reviewer will note.

ML claims I can safely make:

- Every prediction, confidence value, and SHAP explanation traces to a real, fitted model run against real (synthetic) data — nothing hardcoded or invented at serve time.
- The advanced model shows a measured 3.9x lift in Precision@5 over a tabular-only baseline on held-out, temporally-later complaints.
- Point-in-time discipline and complaint-level calibration splitting were both deliberately engineered to prevent specific, named leakage modes, with an automated test guarding the most dangerous one.
- Confidence is a genuine CalibratedClassifierCV output, not a raw score or a UI-side transformation.

ML claims I should NOT make:

- "This model would work on real cybercrime data" — untested, unknowable from this dataset alone.
- "The graph features generalize beyond this generator's specific ring structure" — plausible but not proven.
- "Precision@1 of 17.4% is good" in an absolute sense — say it in lift terms (vs. 4.3% baseline), not as a bare percentage.
- "The model has been cross-validated" — it has not; one seed, one split.

Part 4 — Graph Review

Nodes: accounts (victim/mule/normal — generator metadata, not fed to the model) and withdrawal points (lat/lon/city/bank).

Edges: transfer (account→account) and withdrawal (account→withdrawal point), both carrying amount, timestamp, complaint_id.

Is the graph actually useful for prediction? Yes, verifiably — community size, community complaint count, community geo density, local betweenness, degree centrality, and chain depth are all graph-derived and are all present in the advanced feature set, and removing them (the baseline) measurably drops Precision@5 from 48.6% to 12.3%. This is about as close to "prove the graph matters" as a hackathon project gets.

Is community detection actually used? Yes, directly — feeds community size/complaint-count/geo-density, all real model inputs.

Is centrality actually used? Yes, but its practical contribution is likely small compared to community/geo features — no SHAP importance breakdown is reported anywhere to confirm its weight (a gap).

Is cross-case linking meaningful? Yes — the project's strongest claim (see Part 5).

Are 2-hop/k-hop relationships used? Yes, in two correct ways: the complaint's own traced hop chain feeds `chain_depth`, and the `/graph/{id}` subgraph builder explicitly caps to a curated small neighborhood rather than a naive k-hop BFS — a real, documented fix for a real explosion bug (a shared popular ATM pulling in hundreds of unrelated accounts).

Does the graph reveal something a table-based system would miss? Yes, concretely: two victims sending money to different mule accounts that only turn out to be linked because both forward to a third, shared mule two hops later — no column in a plain complaints table captures this; only graph traversal + community detection surfaces it.

Most important question — is the graph genuinely part of the intelligence, or merely a visualization?

Genuinely part of the intelligence. The measured baseline-vs-advanced gap is the proof: strip the graph features out and Precision@5 falls by more than half. The Fund Flow Trace visualization is a real rendering of the same underlying structure the model consumes — not a separately-drawn decorative diagram disconnected from the prediction path.

Part 5 — Cross-Case Intelligence

Can it take Case $A \rightarrow X \rightarrow Y$ and Case $B \rightarrow Z \rightarrow Y$ and conclude they're connected? Yes — traced exactly:

1. Case A's traced chain ends at some frontier account.
2. That frontier account's Louvain community is looked up — computed on the REAL fund-flow graph (accounts A, X, Y, Z, and everyone else's transactions all in one projection).
3. If X, Y (from A) and Z, Y (from B) all landed in the SAME discovered community — because Y is the structural bridge — the related-complaints computation for A will find B, because B's transactions/withdrawal touch the community's account set, which includes Y.
4. The API goes one step further than "same community": it recomputes the SPECIFIC shared account(s) between A and B, and any shared withdrawal point, and returns them explicitly — so the UI can say "linked via mule account #593766189437" rather than a vague "these are related somehow."

This is genuinely one hop deeper than most "fraud ring" demos, which usually stop at "these accounts are in the same cluster." Here the linkage between any two specific complaints is re-derived and shown with its exact evidence.

SIGNAL	STATUS
Shared accounts	Real, computed
Shared transaction paths	Implicit in the community projection (weighted by transaction count)
Fraud-ring membership	Real, via Louvain — not the generator's ground-truth ring_id
Common cash-out locations	Real, via shared withdrawal points
Temporal similarities	NOT used in cross-case linking — a real gap
Network overlap	Yes — community membership itself is the overlap measure

Is this one of the project's strongest differentiators? Yes, unambiguously. Real graph reasoning producing a concrete, evidenced answer to "why are these connected" — not a keyword match or manual tagging.

Part 6 — Prediction Quality

Candidate generation: every withdrawal point registered in the victim's own city (~40 candidates typically) — not a fixed shortlist, not hand-picked.

Top-5 production: real ranking — score every candidate with the calibrated model, sort descending, take the top 5.

Confidence source: `CalibratedClassifierCV.predict_proba` on the fused feature vector — a real number, not a display artifact. "17%" is the genuine model output rounded for display.

Can it explain "why this location" using actual model evidence? Yes — SHAP `top_features` are computed per-row from the actual fitted tree ensemble, and the narrative sentence references the REAL community complaint count/size for that specific candidate.

SCENARIO	WHAT HAPPENS
No historical match	Narrative explicitly says so ("No prior linked complaints found... ranking is driven by geospatial and recency signal.") — honest fallback, not a silent hallucination.
Account is new / ring unknown	Community-derived features degrade to near-zero gracefully; ranking falls back to geo-density + recency + bank match.
Insufficient transaction history	Raises a dedicated exception, caught by a FastAPI handler returning a clear 422 with a human-readable message — not an unhandled 500.
Multiple locations with similar scores	No explicit "these are close, be careful" signal is surfaced — a real gap.
Model is uncertain (low confidence)	Displayed via lift-over-random framing ("2.9x more likely than random among 40 candidates") rather than a bare, alarming-looking low percentage — a genuinely thoughtful honesty choice.

Part 7 — Explainability Review

"Why did your model choose this ATM?" — can it answer with actual evidence? **Yes**, with real numbers behind every clause: fraud-ring relationship (community complaint count, real), recent transaction (recency decay from the real last-hop timestamp), geographic proximity (real haversine-kernel density), account centrality (real, though weakly weighted in practice), transaction amount (real), previous linked complaints (same community signal, cross-referenced with the related-complaints endpoint for specific IDs).

Gap: the polished narrative sentence is a FIXED TEMPLATE with two numbers substituted in — not generated per-SHAP-value. The raw top_features list (the actual ranked SHAP evidence) IS shown separately and IS real, but the "readable sentence" a judge sees first is one of two canned strings. A sharp judge who sees the SAME sentence structure across two different predictions will correctly suspect it's templated — because it is.

Ideal explanation format (partially achieved): *"Ranked #1 with 17% confidence (2.9x more likely than random among 40 candidates). Primary drivers: this account's cluster has cashed out near this ATM before (community_geo_density=8.8, strongest positive factor); the fund trail reached this account only 40 minutes ago (recency_decay=0.94); 37 other complaints already touch this same 43-account cluster."* Assembled directly from the real top_features list per-row, not picked from 2 canned templates — a concrete, cheap fix.

Part 8 — UI / UX Review

Reviewed as an investigation product, not a SaaS dashboard.

Command Center: 4 stat cards, a hotspots map+list panel, a Live Investigation Feed panel, two more tables below, plus a Simulate Complaint control and arrival-notification card — 5 distinct panels plus a header control on one screen. Individually each is defensible; together it is dashboard clutter by the literal definition — no single panel is unambiguously "the one thing to look at first."

Case page: Good instinct — a persistent header showing the top predicted location + confidence + urgency + Approve/Reject, visible regardless of which tab is open. This was a real usability fix made mid-project (it wasn't the original design — it was originally buried in tab 4). That it needed to be fixed by request, rather than being the first design, is itself informative.

Fund-flow graph: Real data, correctly curated (not BFS explosion), but likely reads as a generic force-directed node cluster without deliberate role-based color coding (victim/mule/frontier), not just community coloring.

Cash-out map: Genuinely good — fitBounds to actual data extent, risk-tiered colored markers, a legend correctly repositioned when it overlapped the map's own labels. One of the more polished screens.

Intervention Brief: Functional, honest about being "not auto-executed," but visually the least distinctive screen.

Typography/color hierarchy: Consistent design tokens, categorical series colors used correctly, status colors reserved for status — better engineered than the median hackathon UI. But the overall visual language is still "dark admin dashboard" — nothing about the chrome itself signals "cybercrime investigation" versus "generic ops console." That signal is entirely in the content, not the visual design language.

Empty/loading/error states: Present and consistent throughout — a real, disciplined pattern, not ad hoc per page.

WHAT SHOULD BE REMOVED	WHAT SHOULD BE ADDED	WHAT SHOULD BE REDESIGNED
Live Feed's undisclosed simulated cadence for the judged demo; merge/remove the two bottom Command Center tables that duplicate dedicated pages	A single unambiguous "most urgent case right now" call-to-action; a score-closeness/uncertainty indicator; role-coded fund-flow graph nodes	Command Center's information hierarchy — every panel currently has roughly equal visual weight

Part 9 — Investigator Experience

Walking through as an investigator when a complaint arrives:

1. **What do I see first?** Command Center — stats, a map, a live feed, tables. No forced entry point.
2. **What do I investigate next?** Click into the complaint from Alerts or Cases — clear enough once I know where to look.
3. **Can I understand the money flow?** Yes, via Fund Flow Trace — real and traceable, though role-coding isn't obviously distinct at a glance.
4. **Can I identify related cases?** Yes, clearly, with real shared-account evidence — one of the best-supported answers in the product.
5. **Can I understand the prediction?** Mostly — Top-5 + confidence + lift framing is clear; no sense of how close #1 is to #2.
6. **Can I see where I should act?** Yes — predicted ATM/branch is prominent, on the map and in the case header.
7. **Can I understand why?** Partially — the evidence is real, but the sentence wrapping it is templated, so it can feel repetitive across cases.
8. **Can I generate an actionable brief?** Yes, and approve/reject from any tab now — genuinely usable.

Confusion points: Command Center's panel count with no forced hierarchy; no visible signal for "this prediction is uncertain" when scores are close; watching the Live Feed for a few minutes might reasonably raise "is this actually real-time?" — a question the UI has no honest built-in answer to.

Part 10 — Demo Review

The 5-Minute Killer Demo

TIME	WHAT TO DO
0:00–0:30	Frame it in one sentence: "Cybercrime complaints get investigated in isolation. We build the money graph, predict where cash gets pulled out, and connect it to the wider ring — automatically." Show Command Center for 5 seconds, no more.
0:30–1:30	Click Simulate Complaint, live. A real held-back complaint reveals, a real predict call runs in front of them. Your strongest "not a slideshow" moment — use it early.
1:30–2:30	Open the case, show the Fund Flow Trace. Point at the traced chain, then the SHAP-backed "why this location" evidence. Say: "this is SHAP output from the actual fitted model, not text we wrote."
2:30–3:30	Cross-case linking — the killer feature. Show a specific shared account number connecting this case to another one, filed by a different victim, in a different city. Say: "the system found this by graph community detection — no one told it these were related."
3:30–4:15	Analytics: show the real baseline-vs-advanced bar chart, 12.3%→48.6% Precision@5. "Held-out test complaints, temporal split, never seen during training." Your only slide-shaped moment — keep it to 30 seconds.
4:15–5:00	Close on the map + Intervention Brief. Predicted ATM on the map, Approve for Action. "This becomes an actual dispatch instruction to the bank's fraud team."

Hide: the Live Investigation Feed's automatic cadence, or narrate it explicitly as simulating ongoing activity between actions if shown at all.

WOW moment: cross-case linking with a named, specific shared account number.

What will make judges suspicious: confidence values that never exceed ~20-25% (explain the lift framing BEFORE they ask); the Live Feed's regular cadence if left running unexplained; SHAP sentences that sound identical across two cases.

Part 11 — What Will Disappoint Me If I Present This?

- **The Live Investigation Feed's cadence is simulated and nothing in the UI says so.** The code is honest about it; the product surface is not. The single biggest gap between what's built and what a user would assume.
- **SHAP narrative sentences are templated, not per-row generated.** The evidence is real; the sentence wrapping it is one of two canned strings.
- **Confidence values topping out around 20-25%** will look weak unless the lift-over-random framing is led with, not offered after the fact.
- **Only one train/test split, one seed reported.** No answer currently exists for "how stable is this number."
- **Cross-case linking has no temporal-similarity signal** despite the product statement implying full pattern-matching.
- **Command Center is visually and informationally dense** without a forced hierarchy — five roughly-equal-weight panels, no single "start here."
- **Entity extraction is regex-over-a-hardcoded-bank-list**, working only because the synthetic narrative templates are trivially regular.

None of these are "actually hardcoded" or "fake" in the sense of fabricated intelligence — I specifically went looking for fabricated SHAP, fake confidence, fake rings, and frontend-only simulation of the CORE pipeline, and did not find them. These are honesty-of-framing and polish gaps, not fraud.

Part 12 — Biggest Differentiators

Implemented, genuinely real

1. Cross-case linking via discovered graph community + specific shared-account evidence.
2. A real, measured baseline-vs-advanced lift (12.3%→48.6% Precision@5) with an automated regression-floor test guarding it.
3. Point-in-time leakage discipline enforced at graph-construction and feature-aggregation level, with an automated test guarding the most dangerous leak.
4. SHAP wired to the actual raw fitted booster, producing real per-row evidence.
5. Lift-over-random-chance confidence framing — honest presentation of modest-looking raw probabilities.

Potential differentiators that are currently only ideas

- Temporal-similarity cross-case linking.
- Per-row-generated (not templated) natural-language SHAP narratives.
- Real-time integration with an actual bank/NCRP system (explicitly out of scope, correctly).
- Cross-validated / multi-seed evaluation stability claims.
- Any drift-monitoring or model-retraining pipeline.

Part 13 — Real-World Feasibility

AREA	ASSESSMENT
Scalability	SQLite, in-process Louvain recompute on a 90s TTL cache — fine for a demo, not for real transaction volume.
Security	RBAC-in-data-filtering only; login is an explicitly-disclosed self-declared persona picker, no passwords, no sessions.
Privacy	Synthetic data only — no real PII exposure risk in the current build.
Data quality	Realistic-shaped but small and template-generated; real NCRP data would be far messier.
False positives	Measured, but the saved report shows zero predictions crossed the 0.5 threshold used — currently uninformative on this data.
Model drift	No monitoring, no retraining trigger.
Auditability	No persisted decision log beyond the in-memory event feed, which resets on restart.
Human oversight	Genuinely present and correctly scoped — Approve/Reject explicitly does not auto-execute anything.
Integration with police/cybercrime systems	None, and none claimed.
Real-time processing	Simulated, not real (see Part 2/11).
Database scalability	SQLite is a demo-scale choice; Postgres/Docker path is written but unverified in this environment (disclosed in README).

Already realistic for a prototype: the ML pipeline discipline (leakage guards, calibration, held-out evaluation), the RBAC-in-data enforcement pattern, the honest "not auto-executed" framing on the one action with real-world consequence.

Part 14 — Competition Judge Attack (20 Hardest Questions)

Q1. Is this real data or synthetic?

WHY ASKED

Table stakes.

ANSWER IN CODE

README/Blueprint Section 14, disclosed openly.

CURRENT ANSWER

Strong — proactively disclosed, not hidden.

Q2. How do you know your model isn't just memorizing your own generator's patterns?

WHY ASKED

Exactly the right skeptical question given the "hot withdrawal points" mechanism.

CODE ANSWER

Temporal split + a distinct unseen_test_patterns generation step for brand-new mule accounts.

CURRENT ANSWER

Partially strong — the unseen-pattern set exists but isn't reported as its own separate evaluation slice.

IDEAL

Report Precision@5 on the unseen-pattern subset alone, separately.

Q3. Your confidence is only 17-20%. Why should I trust that?

CODE ANSWER

Lift framing, real random-chance denominator ($1/n_{\text{candidates}}$).

CURRENT ANSWER

Strong if led with, weak if reactive.

Q4. Is your SHAP explanation actually connected to the model, or is it text you wrote?

CODE ANSWER

TreeExplainer on the real fitted booster; but the sentence is templated.

CURRENT ANSWER

Strong on the evidence, honest weakness on the sentence — say so yourself.

Q5. Why is your baseline so weak (4.3% Precision@1)? Did you cripple it to make your model look better?

CODE ANSWER

Baseline is deliberately deprived of graph/temporal features BY DESIGN, to isolate the graph's contribution.

CURRENT ANSWER

Strong — a legitimate ablation, not a strawman.

Q6. Is your cross-case linking just "same community" or does it show real evidence?

CODE ANSWER

Specific shared account IDs, not just a community number.

CURRENT ANSWER

Strong.

Q7. What's your false-positive rate?

CODE ANSWER

n_high_confidence: 0 in the actual report at the 0.5 threshold.

CURRENT ANSWER

Weak — report the metric at a lower, more realistic threshold instead.

Q8. Did you cross-validate, or is this one lucky split?

CODE ANSWER

One 80/20 temporal split, one seed.

CURRENT ANSWER

Weak — report mean±std across ≥3 seeds before the next demo.

Q9. Is your Live Feed real-time?

CODE ANSWER

Real data, simulated cadence, explicitly documented in code.

CURRENT ANSWER

Weak if asked directly, unless volunteered first.

Q10. What happens with real, messy narrative text instead of your templates?

CODE ANSWER

Regex/hardcoded bank list, admittedly narrow.

CURRENT ANSWER

Honest weakness — own it directly.

Q11. Does the graph actually change your prediction, or is it just for the visualization?

CODE ANSWER

Measured Precision@5 lift when graph features are included.

CURRENT ANSWER

Strong — point directly at the report numbers.

Q12. Could two very different-looking cases get linked incorrectly just because they share a popular ATM?

CODE ANSWER

Withdrawal-point nodes are deliberately excluded from the Louvain projection specifically to prevent this.

CURRENT ANSWER

Strong — a real, deliberate design decision.

Q13. What's your model's mean rank of the true point?

CODE ANSWER

10.4 (of ~40 candidates) advanced vs 18.8 baseline.

CURRENT ANSWER

Moderate — present honestly as "top quartile," not "always top-5."

Q14. How would this scale to millions of complaints?

CODE ANSWER

SQLite + per-request Louvain recompute, cached with a 90s TTL.

CURRENT ANSWER

Weak — disclose plainly that batch graph computation would replace per-request recompute at real scale.

Q15. Is your RBAC real security?

CODE ANSWER

City-scoping enforced server-side on every endpoint; persona selection is self-declared, no auth.

CURRENT ANSWER

Honest and correctly scoped — say exactly this.

Q16. What stops an investigator approving a low-confidence, likely-wrong prediction?

CODE ANSWER

Nothing currently.

CURRENT ANSWER

Weak, real gap — a confidence-based warning before approval is a cheap, high-value fix.

Q17. Do you detect NEW fraud rings the generator didn't design, or only the ones it built in?

CODE ANSWER

Louvain runs on the full real graph including background noise, so in principle yes — but every ring in the dataset is generator-designed.

CURRENT ANSWER

Honestly can't fully answer — say so.

Q18. Why XGBoost and not a graph neural network?

CODE ANSWER

Deliberate scope decision.

CURRENT ANSWER

Strong — dataset size doesn't support a GNN without overfitting.

Q19. If I restart your backend, does anything important get lost?

CODE ANSWER

Live Feed history and settings toggle, yes — all in-memory. Trained models and DB survive.

CURRENT ANSWER

Be upfront, low stakes but real.

Q20. Can you show me the raw evaluation JSON, not just the chart?

CODE ANSWER

Yes, a real file, served verbatim by the evaluation endpoint.

CURRENT ANSWER

Strong — have it open in a terminal tab, ready to show.

Part 15 — Priority Fixes (Top 10)

P0 — Must Fix Before Demo

#	PROBLEM	WHY IT MATTERS	EXACT CHANGE	IMPACT	DIFFICULTY
1	Live Feed's simulated cadence is undisclosed in the UI	Single biggest "gotcha" risk if asked directly	Add a small honest label, or don't run it during the judged demo window	Removes largest credibility risk	Trivial
2	No answer for "one split, one seed"	Direct ML rigor attack, currently unaddressed	Re-run evaluate.py with 3-5 seeds, report mean $\pm$ std	Converts a weakness into a strength, near-zero effort	Trivial
3	False-positive metric reports n=0	"What's your FP rate" is a near-certain question; zero samples sounds evasive	Lower the confidence threshold in evaluate.py to one the distribution actually reaches, rerun	Turns a dead metric into a real one	Trivial

P1 — High Value

#	PROBLEM	WHY IT MATTERS	EXACT CHANGE	IMPACT	DIFFICULTY
4	SHAP narratives are templated, not per-row generated	Repeats verbatim across cases, undermining the "real explainability" claim on inspection	Build the sentence from the actual top 2-3 SHAP feature/value pairs already computed	Closes the single most inspectable "is this real" gap	Medium
5	No uncertainty signal on near-tied candidates	False confidence is a real operational risk, and an easy judge question	Show a "closely contested — verify before acting" note when the score gap is small	Improves both investigator trust and judge perception	Low
6	Command Center has 5+ equal-weight panels	"What should I look at first" is unanswered	Demote/merge the two bottom tables that duplicate dedicated pages	Meaningfully improves demo clarity, no functionality lost	Low
7	No temporal-similarity signal in cross-case linking	Implied by the product's own pitch, not delivered	Add a "cached out within X hours of each other" flag using timestamps already available	Closes a real gap in the strongest differentiator	Low-medium

P2 — Nice to Have

#	PROBLEM	WHY IT MATTERS	EXACT CHANGE	IMPACT	DIFFICULTY
8	Entity extraction is a hardcoded 6-bank regex list	Doesn't generalize, but already honestly scoped	None required before demo; own it directly if asked	Low urgency	N/A for now
9	No confidence-based gate on Approve/Reject	A real investigator system shouldn't silently approve a low-confidence prediction	Reuse fix #5's near-tie/low-confidence warning next to the Approve button	Moderate usability/credibility improvement	Low

P3 — Future

#	PROBLEM	WHY IT MATTERS	EXACT CHANGE	IMPACT	DIFFICULTY
10	No multi-run stability reporting, no unseen-ring ablation reported separately, no drift/audit infrastructure	Genuinely important for real deployment, not Po for a department-level demo	None needed now; mention as a "next steps" roadmap if asked	Signals maturity of thinking without the engineering	N/A now

Final Output

1. Overall Score

74 / 100.

2. What We Built Well

A real, leakage-guarded, point-in-time-disciplined ML pipeline with a genuine, measured, regression-tested baseline-vs-advanced lift (12.3%→48.6% Precision@5); correctly-wired SHAP on the raw fitted model; real, evidenced cross-case linking through discovered graph structure, not a lookup table; an honest calibration story (complaint-level split, a documented isotonic-collapse bug found and fixed); and an unusually rigorous test suite for a project at this level.

3. What Is Actually Intelligent

Graph construction, Louvain community detection, cross-case linking, temporal decay, geographic density, XGBoost ranking, calibration, and SHAP explanation are all REAL COMPUTATION, traced to specific executed code, verified against a real stored evaluation report — not simulated, not hardcoded, not decorative.

4. What Is Weak / Fake / Superficial

Entity extraction is regex-over-a-hardcoded-list, not NLP (disclosed honestly in-code). SHAP's underlying evidence is real, but its narrative sentence is templated, not per-row generated. The Live Investigation Feed's cadence is simulated with no in-UI disclosure. Only one train/test split and seed has been evaluated. The false-positive-rate metric currently has zero qualifying samples.

5. Biggest Technical Risk

A hostile judge asking "how stable is your 48.6% number across different seeds/splits" and getting no prepared answer — a five-minute fix that hasn't been done.

6. Biggest UI/UX Problem

Command Center has no forced visual hierarchy across its 5+ panels — nothing tells an investigator where to look first, undercutting the product's own core promise of "understand what's happening" at a glance.

7. Biggest Competition Risk

The Live Investigation Feed reading as more "real-time" than it honestly is, if a judge notices its regular cadence and asks directly, with no framing offered first.

8. Top 5 Differentiators

Cross-case linking with named shared-account evidence; a real measured baseline-vs-advanced lift with a regression-tested floor; point-in-time leakage discipline with an automated ground-truth-leak guard; SHAP wired to the actual raw booster; the lift-over-random-chance confidence framing.

9. Top 10 Changes Before Demo

See Part 15, in order: (1) disclose or hide the Live Feed's simulated cadence, (2) rerun evaluation across multiple seeds and have stability numbers ready, (3) fix the false-positive-rate metric's threshold, (4) generate SHAP narratives per-row instead of from templates, (5) add a near-tie/uncertainty flag, (6) simplify Command Center's panel hierarchy, (7) add temporal-similarity to cross-case linking, (8–10) lower priority per Part 15.

10. Final Verdict

ALMOST READY.

The ML core is real, defensible, and would survive a genuinely skeptical technical cross-examination on almost every point except split/seed stability and the false-positive metric — both of which are re-run-an-existing-script fixes, not rebuilds. The product's biggest exposure isn't fake intelligence (there isn't any to find here) — it's a UI that oversells the Live Feed's real-time-ness by omission, and a Command Center that hasn't had a final "what's the one thing to look at" edit pass. Fix the Pos (all cheap, all achievable in under a day), rehearse the P1 talking points (all things provable, just not yet said out loud), and this is a genuinely strong department-level submission — not because it avoided hard questions, but because most of the hard questions already have real, correct answers sitting in the code, just not yet in your mouth.

Generated by Claude Sonnet 5 · Strict technical evaluation of the PredicTrace codebase as of this review · Every claim traced to a specific file and line, not inferred from naming or documentation alone.